

chain rule

$$y = f(ax+b)$$

$$\Rightarrow \frac{dy}{dx} = af'(ax+b)$$

$$y = [f(x)]^n$$

$$\Rightarrow \frac{dy}{dx} = n [f(x)]^{n-1} f'(x)$$

when $n \in \mathbb{R}$.

$$y = x^n$$

$$\Rightarrow y' = nx^{n-1}$$

$$y = \sqrt{x}$$

$$\Rightarrow y' = \frac{1}{2\sqrt{x}}$$

$$y = \sqrt{f(x)}$$

$$\Rightarrow y' = \frac{f'(x)}{2\sqrt{f(x)}}$$